Math 526, F08
Exam 2

Name: Answer
SSN: xxx-xx-

Instructions:

- There are totally 6 problems.
- You must show all of your work to receive a credit for a correct answer.
- No graphical calculator is allowed in the quiz and exam.

Problem	Points	Score
1	20	
2	15	
3	15	
4	15	
5	20	
6	15	
Total	100	

Good Luck !

Problem 1. (20 points) Let $\mathbf{A} = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 8 & 10 \\ 3 & 6 & 7 & 10 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} 0 \\ 2 \\ -2 \end{bmatrix}$.

(a) Reduce $[\mathbf{A}\ \ \mathbf{b}]$ to $[\mathbf{R}\ \ \mathbf{d}]$ where $\mathbf{R}$ denotes the reduced echelon form of $\mathbf{A}$.

Solution:

$$[A\ b] = \left[\begin{array}{cccc:c} 1 & 2 & 3 & 4 & 0 \\ 2 & 4 & 8 & 10 & 2 \\ 3 & 6 & 7 & 10 & -2 \end{array}\right] \to \left[\begin{array}{cccc:c} 1 & 2 & 3 & 4 & 0 \\ 0 & 0 & 2 & 2 & 2 \\ 0 & 0 & -2 & -2 & -2 \end{array}\right]$$

$$\to \left[\begin{array}{cccc:c} 1 & 2 & 3 & 4 & 0 \\ 0 & 0 & 2 & 2 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right] \to \left[\begin{array}{cccc:c} 1 & 2 & 3 & 4 & 0 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

$$\to \left[\begin{array}{cccc:c} 1 & 2 & 0 & 1 & -3 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right] = [R\ d]$$

(b) Find a particular solution $\mathbf{x}_p$ of $\mathbf{Ax} = \mathbf{b}$ by setting the free variables to be zero.

Solution: Setting $x_2 = x_4 = 0$, we get

$$x_1 = -3, \quad x_3 = 1$$

So that $x_p = \begin{bmatrix} -3 \\ 0 \\ 1 \\ 0 \end{bmatrix}$ is a particular solution.

(c) Find the complete solution to $\mathbf{Ax} = \mathbf{b}$.

Solution: setting $x_2 = 1$, $x_4 = 0$, we get $(Rx=0)$

$$\begin{array}{l} x_1 + 2x_2 + x_4 = 0 \\ x_3 + x_4 = 0 \end{array} \Rightarrow \begin{array}{l} x_1 = -2 \\ x_3 = 0 \end{array} \Rightarrow s_1 = \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix}$$

setting $x_2 = 0$, $x_4 = 1$, we get

$$\begin{array}{l} x_1 = -1 \\ x_3 = -1 \end{array} \Rightarrow s_2 = \begin{bmatrix} -1 \\ 0 \\ -1 \\ 1 \end{bmatrix}$$

So that $x = x_p + c_1 s_1 + c_2 s_2 = \begin{bmatrix} -3 \\ 0 \\ 1 \\ 0 \end{bmatrix} + c_1 \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} -1 \\ 0 \\ -1 \\ 1 \end{bmatrix}$

Problem 2. (15 points)

Let $\mathbf{A} = \begin{bmatrix} 2 & 4 & 0 & 5 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$.

(a) Find the dimension and a basis for the row space of $\mathbf{A}$, $C(\mathbf{A}^T)$.

$\dim(C(A^T)) = 2$,

One basis : $\begin{bmatrix} 2 \\ 4 \\ 0 \\ 5 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \\ 2 \end{bmatrix}$

(b) Find the dimension and a basis for the column space of $\mathbf{A}$, $C(\mathbf{A})$.

$\dim(C(A)) = 2$

One basis : $\begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$

(c) Find the dimension and a basis for the nullspace of $\mathbf{A}$, $N(\mathbf{A})$.

$\dim(N(A)) = 4 - 2 = 2$

$Ax = 0 \Rightarrow 2x_1 + 4x_2 + 5x_4 = 0$

$x_3 + 2x_4 = 0$

setting $x_2 = 1, x_4 = 0 \Rightarrow \begin{cases} x_1 = -2 \\ x_3 = 0 \end{cases} \Rightarrow s_1 = \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix}$

setting $x_2 = 0, x_4 = 1 \Rightarrow \begin{cases} x_1 = -5/2 \\ x_3 = -2 \end{cases} \Rightarrow s_2 = \begin{bmatrix} -5/2 \\ 0 \\ -2 \\ 1 \end{bmatrix}$

One basis : $\begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -5/2 \\ 0 \\ -2 \\ 1 \end{bmatrix}$

(d) Find the dimension and a basis for the left nullspace pf $\mathbf{A}$, $N(\mathbf{A}^T)$.

$\dim(N(A^T)) = 3 - 2 = 1$

$A^T y = 0 \Rightarrow y_1 \begin{bmatrix} 2 \\ 4 \\ 0 \\ 5 \end{bmatrix} + y_2 \begin{bmatrix} 0 \\ 0 \\ 1 \\ 2 \end{bmatrix} = 0 \Rightarrow y_1 = y_2 = 0$

$y(y_1, y_2, y_3)$

one basis : $\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$

Problem 3. (15 points) Find the straight line $C + Dt$ that comes closest to the values $\mathbf{b} = 0, 8, 8, 20$ at times $t = 0, 1, 3, 4$ by the least squares fitting. (Hint: set up and solve $\mathbf{A}^T\mathbf{A}\hat{\mathbf{x}} = \mathbf{A}^T\mathbf{b}$)

Solution: $A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 3 \\ 1 & 4 \end{bmatrix}$, $\hat{x} = \begin{bmatrix} C \\ D \end{bmatrix}$, $b = \begin{bmatrix} 0 \\ 8 \\ 8 \\ 20 \end{bmatrix}$

$$A^TA = \begin{bmatrix} 4 & 8 \\ 8 & 26 \end{bmatrix}, \quad A^Tb = \begin{bmatrix} 36 \\ 112 \end{bmatrix}$$

$$\begin{bmatrix} 4 & 8 \\ 8 & 26 \end{bmatrix}\begin{bmatrix} C \\ D \end{bmatrix} = \begin{bmatrix} 36 \\ 112 \end{bmatrix}$$

$$\Rightarrow C = 1 \quad D = 4$$

$$\Rightarrow y(t) = 1 + 4t.$$

Problem 4. (15 points) Let $\mathbf{A} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & 0 \\ 2 & 0 & 4 \end{bmatrix}$. Find an orthonormal basis $\mathbf{q}_1$, $\mathbf{q}_2$, $\mathbf{q}_3$ for the column space of $\mathbf{A}$ using the **Gram-Schmidt** process.

Solution:

$$a_1 = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}, \quad a_2 = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \quad a_3 = \begin{bmatrix} 1 \\ 0 \\ 4 \end{bmatrix}$$

$$\tilde{q}_1 = a_1 = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}, \qquad \|\tilde{q}_1\|^2 = 6$$

$$\tilde{q}_2 = a_2 - \frac{\tilde{q}_1^T a_2}{\|\tilde{q}_1\|^2} \tilde{q}_1$$

$$= \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} - \left(\frac{0}{6}\right) \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} \qquad \|\tilde{q}_2\|^2 = 2$$

$$\tilde{q}_3 = a_3 - \frac{\tilde{q}_1^T a_3}{\|\tilde{q}_1\|^2} \tilde{q}_1 - \frac{\tilde{q}_2^T a_3}{\|\tilde{q}_2\|^2} \tilde{q}_2$$

$$= \begin{bmatrix} 1 \\ 0 \\ 4 \end{bmatrix} - \left(\frac{9}{6}\right) \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} - \left(\frac{1}{2}\right) \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \\ 0 \\ 4 \end{bmatrix} - \begin{bmatrix} 3/2 \\ 3/2 \\ 3 \end{bmatrix} - \begin{bmatrix} 1/2 \\ -1/2 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix} \qquad \|\tilde{q}_3\|^2 = 3$$

$$\Rightarrow q_1 = \frac{\tilde{q}_1}{\|\tilde{q}_1\|} = \begin{bmatrix} 1/\sqrt{6} \\ 1/\sqrt{6} \\ 2/\sqrt{6} \end{bmatrix}, \quad q_2 = \frac{\tilde{q}_2}{\|\tilde{q}_2\|} = \begin{bmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \\ 0 \end{bmatrix}$$

$$q_3 = \frac{\tilde{q}_3}{\|\tilde{q}_3\|} = \begin{bmatrix} -1/\sqrt{3} \\ -1/\sqrt{3} \\ 1/\sqrt{3} \end{bmatrix}$$

Problem 5. (20 points) Let $\mathbf{A} = \begin{bmatrix} 0 & 1 & 3 \\ 1 & 0 & 1 \\ 2 & 1 & 1 \end{bmatrix}$.

(a) Solve the linear system $\mathbf{Ax} = \mathbf{b}$ with $\mathbf{b} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ by **Cramer's Rule**.

Solution: $\det A = (-1)\begin{vmatrix} 1 & 1 \\ 2 & 1 \end{vmatrix} + 3\cdot\begin{vmatrix} 1 & 0 \\ 2 & 1 \end{vmatrix} = 1+3=4$

$$B_1 = \begin{bmatrix} 1 & 1 & 3 \\ 0 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix} \Rightarrow \det B_1 = -1$$

$$B_2 = \begin{bmatrix} 0 & 1 & 3 \\ 1 & 0 & 1 \\ 2 & 0 & 1 \end{bmatrix} \Rightarrow \det B_2 = 1$$

$$B_3 = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 0 \\ 2 & 1 & 0 \end{bmatrix} \Rightarrow \det B_3 = 1$$

$$\Rightarrow x = \begin{bmatrix} \det B_1/\det A \\ \det B_2/\det A \\ \det B_3/\det A \end{bmatrix} = \begin{bmatrix} -1/4 \\ 1/4 \\ 1/4 \end{bmatrix}$$

(b) Find $\mathbf{A}^{-1}$ using the cofactor formula $\mathbf{C}^T/det\mathbf{A}$.

Solution: $C = \begin{bmatrix} -1 & 1 & 1 \\ 2 & -6 & 2 \\ 1 & 3 & -1 \end{bmatrix}$

$$\Rightarrow A^{-1} = \frac{1}{4}\begin{bmatrix} -1 & 2 & 1 \\ 1 & -6 & 3 \\ 1 & 2 & -1 \end{bmatrix}$$

Problem 6. (15 points) Ture or False (no explaination needed).

(a) If the columns of a matrix are independent, so are the rows.

False

(b) The nullspace $N(\mathbf{A})$ and the row space $C(\mathbf{A}^T)$ are orthogonal complements.

True

(c) $\begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} = \begin{vmatrix} -d & e & f \\ -a & b & c \\ -g & h & i \end{vmatrix}$.

True

(d) The area of the triangle formed by the three points $(-1,2)$, $(0,5)$, $(3,3)$ is $\frac{1}{2}|det\mathbf{A}|$ with $\mathbf{A} = \begin{bmatrix} -1 & 2 & 1 \\ 0 & 5 & 1 \\ 3 & 3 & 1 \end{bmatrix}$.

True

(e) $(\mathbf{u} \times \mathbf{v}) \cdot \mathbf{w} = (\mathbf{v} \times \mathbf{u}) \cdot \mathbf{w}$.

False